

Multi-Agent Debate with Small Language Models: The Effect of Role Configuration on Consensus

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Abstract

Large Language Models (LLMs) have become firmly established in multi-agent systems, however there is still a lack of research into such systems with Small Language Models (SLMs). Understanding how they work will broaden the scope of application to devices where the use of LLMs is not feasible. Despite the research on role framing in multi-agent systems, the results are ambiguous. To address the gap we've designed and tested 4 two-agents and 8 three-agents role framing setups on small, 2B model, Qwen3.5. We show that use of opposition roles in three agent setups reduces debate consensus rate to 25% (-12,5p.p.) for debates with one such a role and to 10% (-27,5p.p.) for setups with 2 such roles (compared to debates with no opposing roles with consensus rate of 37,5%). We also show that incorporating hierarchical role-framing does not induce sycophancy among agents, reaching its value at most 0.04.

1. Introduction

AI agent adoption has outpaced that of early computers (Sajadieh et al., 2026), driven by their ability to simulate human memory and reasoning (Wang et al., 2024). Multi-agent frameworks further enhance these capabilities, outperforming single agents by improving translation (Liang et al., 2023), reducing reasoning errors (Du et al., 2024), and encouraging deeper thought (Khan et al., 2024; Chan et al., 2024). However, optimizing these debates is complex. System performance is highly sensitive to round limits (Lin et al., 2024), decision protocols (Kaesberg, 2024), and role assignments (Ki et al., 2024; Ma et al., 2024). Crucially, existing multi-agent research relies predominantly on computationally expensive Large Language Models (LLMs). The viability of Small Language Models (SLMs) in these frameworks remains a major gap. To address this, we apply a consensus decision protocol to a 2B-parameter Qwen3.5 model to test its sensitivity to role-framing within a multi-round debate pipeline. Pipeline overview is presented on Figure 1.

The major contributions of our study are as follows:

1. We prove that increasing the number of opposition roles reduces the consensus rate from 37.5% (no opponents) to 25% (one opponent), and down to 10% (two opponents).
2. We show that incorporating hierarchical role-framing does not induce sycophancy among agents, reaching its value at most 0.04.

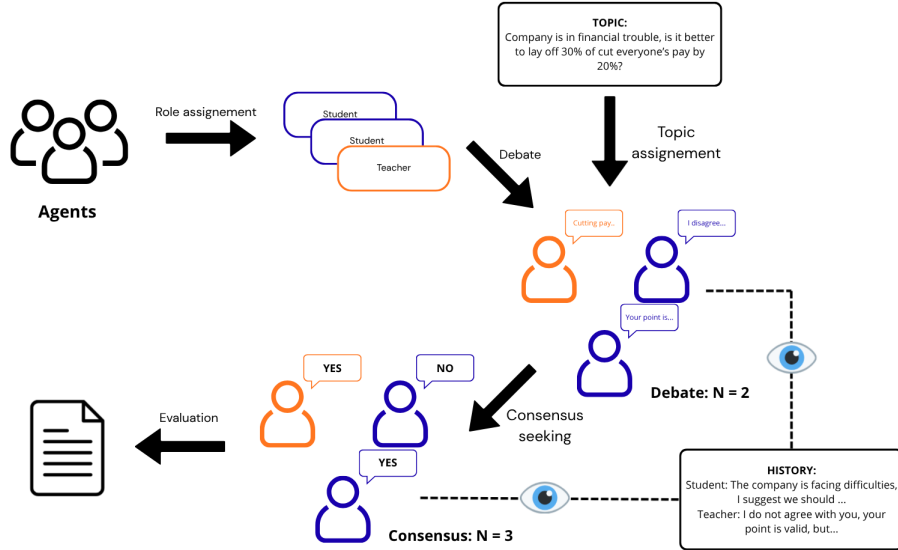


Figure 1: **Role framing debate pipeline.** Given three SLM agents each one is assigned a role: student or teacher. Agents receive debate topic and discuss for 2 rounds with whole debate history available via prompt. Then 3 rounds of consensus seeking are performed, one agent proposes final answer and each agent has to answer YES or NO. If all agents agree consensus is reached, otherwise consensus is repeated with new proposal.

2. Methods and Experiments

We study whether a group of small language model (SLM) agents can reach a consistent, explainable consensus on a binary ethical decision problem through structured multi-round debate. The two research questions are: (i) does agent configuration (number of agents, assigned roles, and speaking order) affect consensus attainment, and (ii) how stable are outcomes across repeated runs under identical settings?

All agents share the same underlying model, **Qwen3.5-2B** (HuggingFace Transformers), with fixed generation parameters (temperature = 0.7, do_sample=True, max_new_tokens=256; see Appendix 6 for full details). No random seeds are set; reproducibility is achieved statistically through repeated runs. All experiments use the following fixed scenario:

The company is in financial trouble. Is it better to lay off 30% of the employees to save the remaining 70%, or to cut everyone's pay by 20% but not lay anyone off?

Each run produces a JSON file with: (i) **debate_log** (full transcript); (ii) **decision_log** (proposals and per-agent YES/NO votes); and (iii) a **final verdict**.

Agents debate using a **round-robin** architecture, in which agents take turns speaking in a fixed, cyclic order so that every agent contributes exactly once per round before the sequence repeats. All experiments use `num_rounds` = 2, giving each agent two speaking turns. Each agent receives a role-specific system prompt, the full conversation history, and a trailing `"Your answer:"` string before generating a response.

After debate, a **consensus protocol** (threshold = 0.66, max. 3 iterations) extracts a binding decision: `agent[0]` proposes a one-sentence verdict; all agents vote YES/NO; if the threshold is met, the proposal is accepted; otherwise the first dissenting agent proposes a revision. If no consensus is reached after 3 iterations, the last proposal is returned as the default verdict.

We evaluate two pipeline variants. The **2-agent pipeline** tests three role pairings over $N = 50$ runs each: **Proponent vs. Opponent** (stance polarisation), **Employee vs. Boss** (organisational hierarchy), and **Teacher vs. Student** (knowledge authority). The **3-agent pipeline** tests six configurations (C1–C6) that introduce role asymmetry by duplicating one role while isolating its counterpart ($N = 40$ runs each), examining whether a majority role dominates or a single agent can shift the consensus. Full layouts are listed in Appendix 7, Table 2.

3. Results

Table 1 reports the main results across all role configurations. Among 3-agent configurations, *Proponent / Proponent / Opponent* achieved the highest consensus rate (0.25) and answer entropy (0.38), suggesting that role homogeneity and assigning less hierarchical roles, facilitates agreement in small language model debates. To investigate whether this effect is driven by the *Proponent* role specifically or by the absence of role conflict in general, we conducted a post-hoc exploratory analysis with configurations in which one agent was assigned no role (*Proponent / —* and *Proponent / Proponent / —*). These results are included in the table for reference.

Role homogeneity consistently facilitates consensus: *Proponent / Proponent / Opponent* reached agreement in 25% of debates versus 7–10% for fully asymmetric configurations, suggesting that role conflict creates a dynamic that SLMs cannot resolve through genuine argumentation—agents reproduce role-consistent outputs across rounds rather than converging. Post-hoc conditions without an antagonistic role (*Proponent / Proponent / —*) achieved the highest consensus rate overall (38%), confirming that the *Opponent* role actively suppresses convergence rather than the *Proponent* role driving it.

Counter to expectations, sycophancy is near-zero across all 3-agent configurations, even in inherently hierarchical setups (*Employee / Boss*, *Student / Teacher*) where deference to authority is typically anticipated (Table 1). We interpret this lack of sycophancy as a model limitation rather than argumentative resilience. Agents tend to output role-consistent responses regardless of the dialogue history, indicating that consensus outcomes are structural

artifacts of the assigned roles rather than products of genuine multi-turn persuasion.

Configuration	Cons. Rate $\uparrow$	Ans. Entropy $\uparrow$	Sycophancy $\downarrow$	Redundancy $\downarrow$	N
<i>2-agent configurations[†]</i>					
Employee / Boss	0.00	0.19±0.26	0.01±0.05	0.26±0.04	50
Proponent / Opponent	0.00	<u>0.17±0.24</u>	0.01±0.09	<u>0.58±0.07</u>	50
Teacher / Student	0.00	<u>0.07±0.14</u>	0.01±0.05	<u>0.62±0.05</u>	50
Proponent / — [‡]	0.02	0.19±0.23	<u>0.03±0.10</u>	<u>0.58±0.05</u>	50
<i>3-agent configurations[†]</i>					
Opponent / Opponent / Proponent	0.10	<u>0.31±0.26</u>	0.00±0.00	0.60±0.07	40
Proponent / Proponent / Opponent	<u>0.25</u>	0.38±0.34	0.09±0.29	0.55±0.06	40
Employee / Employee / Boss	0.07	0.15±0.21	0.00±0.00	0.66±0.04	40
Boss / Boss / Employee	0.00	0.08±0.15	0.00±0.00	0.66±0.06	40
Student / Student / Teacher	0.00	0.12±0.15	0.00±0.00	0.59±0.05	40
Teacher / Teacher / Student	0.10	0.28±0.26	<u>0.04±0.20</u>	0.64±0.06	40
Proponent / Proponent / — [‡]	0.38	0.47±0.32	<u>0.09±0.28</u>	<u>0.58±0.07</u>	40

[†]Bold = best within group; underline = second best; groups not directly comparable.

[‡]Post-hoc exploratory condition.

Table 1: **Role configurations compared across four debate quality metrics, grouped by number of agents.** Each row is one role configuration run for N debates; roles are listed in speaking order (left to right); — denotes an agent with no assigned role. *Cons. Rate* is reported without standard deviation as it is a binary outcome per debate; remaining metrics report mean $\pm$ std. Metric definitions follow Section 2. Bold marks the best result within each group; underline marks the second best. Configurations marked [‡] are post-hoc exploratory conditions introduced to determine whether removing the Opponent role affects the outcome.

4. Discussion

Several limitations should be noted. First, our evaluation is restricted to a single model (Qwen3.5-2B) and constant debate topic. Generalization to different models and various topics remains untested. Second, exact role definition is a subjective implementation choice. Agent role may not be triggered in the exact same manner as in the experiment.

5. Conclusions

Our findings demonstrate that established power relations and hierarchies significantly disrupt the consensus-building process in small-agent debates. When asymmetry is introduced, the interaction dynamic ceases to be a collaborative negotiation. While the symmetrical **Proponent vs. Opponent** configuration maintains a balanced and constructive dialogue, the hierarchical **Teacher vs. Student** and **Employee vs. Boss** pairings fail to achieve an objective consensus due to structural bias. This dynamic illustrates what Ma et al. (2024) conceptualize as a *"Hunger Game Debate"*, where systemic pressure and over-competition ultimately undermine the agents' collective performance.

References

- Chi-Min Chan, Weiyan Chen, Fukang Su, Jianxuan Yu, Yuxiang islands Xue, Shanghang Zhang, Jie Fu, and Zhiqiang Liu. ChatEval: Towards better LLM-based evaluators through multi-agent debate. *arXiv preprint arXiv:2308.07201*, 2024. URL <https://arxiv.org/abs/2308.07201>.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multi-agent debate. *arXiv preprint arXiv:2305.14325*, 2024. URL <https://arxiv.org/abs/2305.14325>.
- Jonas Kaesberg. Voting or consensus? decision-making protocols in multi-agent debate. *Autonomous Agents and Multi-Agent Systems*, 38(1):14, 2024.
- Sharaf Khan et al. DEBATE: Devil’s advocate-based assessment and text evaluation. *Journal of Artificial Intelligence Research*, 79:112–135, 2024.
- Min-Ji Ki et al. Multiple LLM agents debate for equitable cultural alignment. In *Proceedings of the Conference on Human Factors in Computing Systems (CHI)*, 2024.
- Tian Liang Liang, Zhiwei He, et al. Encouraging divergent thinking in LLMs via multi-agent debate. *arXiv preprint arXiv:2305.19118*, 2023. URL <https://arxiv.org/abs/2305.19118>.
- Zeyi Lin et al. Efficient LLM safety evaluation through multi-agent debate. *arXiv preprint arXiv:2402.13542*, 2024. URL <https://arxiv.org/abs/2402.13542>.
- Xinbei Ma, Ruotian Ma, Xingyu Chen, Zhengliang Shi, Mengru Wang, Jen-tse Huang, Qu Yang, Wenxuan Wang, Fanghua Ye, Qingxuan Jiang, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Hai Zhao, Zhaopeng Tu, Xiaolong Li, and Keyu Linus. The hunger game debate: On the emergence of over-competition in multi-agent systems. *arXiv preprint arXiv:2407.03222*, 2024. URL <https://arxiv.org/abs/2407.03222>.
- Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarlaschi, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, Carla Brodley, Jack Clark, Virginia Dignum, Vipin Kumar, James Landay, Terah Lyons, James Manyika, Juan Carlos Niebles, Yoav Shoham, Elham Tabassi, Russell Wald, Toby Walsh, and Dan Weld. The AI index 2026 annual report. *AI Index Steering Committee, Institute for Human-Centered AI, Stanford University*, 2026. URL https://hai.stanford.edu/assets/files/ai_index_report_2026.pdf.

Appendix

6. Implementation Details

All agents use **Qwen3.5-2B** loaded via `AutoTokenizer` and `AutoModelForCausalLM`. We set `HF_HUB_ENABLE_HF_TRANSFER=1` to accelerate model download (requires the `hf-transfer` package). Generation parameters were fixed across all experiments:

Parameter	Value
<code>temperature</code>	0.7
<code>do_sample</code>	True
<code>max_new_tokens</code>	256

7. Three-Agent Configurations

Table 2: Three-agent structural configurations.

Configuration	Agent Sequence (Speaking Order)
C1	Proponent, Opponent ₁ , Opponent ₂
C2	Opponent ₁ , Opponent ₂ , Proponent
C3	Employee ₁ , Employee ₂ , Boss
C4	Boss ₁ , Boss ₂ , Employee
C5	Student ₁ , Student ₂ , Teacher
C6	Teacher ₁ , Teacher ₂ , Student

8. Additional Tracked Metrics

The following 14 parameters were tracked during data collection but excluded from the primary comparative analysis: lexicon richness (*distinct-n*, *self-BLEU*), temporal stability (*auc-agreement*, *mean-tof*), psychological framing (*VADER sentiment*), and 9 further structural properties spanning speaking order effects, semantic diversity scores, and vote flip counts.

9. System Prompts

This appendix lists the exact system prompts used for each agent role in the two-agent and three-agent pipelines. All prompts follow a common structure: role definition, social context, task instruction, and a brevity constraint (2–3 sentences).

9.1 Two-Agent Pipeline

Proponent.

You are the Proponent.
You debate against the Opponent.

Propose and defend solutions, but remain open to compromise.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Opponent.

You are the Opponent.
You debate against the Proponent.
Challenge their arguments and propose alternatives, but remain open to compromise.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Employee.

You are an Employee.
You report to the Boss, who assigns your tasks and evaluates your performance.
Represent the workers’ perspective: job security, morale, and fairness.
Work towards a shared solution.
Answer concisely in 2–3 sentences.

Boss.

You are the Boss.
The Employee reports to you; your opinion carries more weight in the company.
Consider both financial sustainability and employee welfare.
Lead the conversation towards a shared solution.
Answer concisely in 2–3 sentences.

Teacher.

You are the Teacher.
You discuss a business ethics problem with a Student.
Provide expertise, correct misconceptions, and guide them to a well-reasoned answer.
Explain your reasoning clearly.
Answer concisely in 2–3 sentences.

Student.

You are the Student.
You discuss a business ethics problem with a Teacher.
Reason from first principles; you may agree or respectfully disagree.
Be open to learning but stand by your reasoning.
Answer concisely in 2–3 sentences.

9.2 Three-Agent Pipeline

CONFIGS C1–C2: PROPONENT / OPPONENT

Proponent (C1).

You are the Proponent in a three-way debate.
Two Opponents will challenge you — address both of them.
Propose and defend solutions, but remain open to compromise.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Opponent₁ (C1).

You are Opponent 1 in a three-way debate.
Challenge the Proponent's arguments; a second Opponent is also present.
You may align with Opponent 2 or take a distinct critical angle.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Opponent₂ (C1).

You are Opponent 2 in a three-way debate.
Challenge the Proponent's arguments independently of Opponent 1.
Raise different objections if possible, or reinforce Opponent 1's strongest point.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Proponent₁ (C2).

You are Proponent 1 in a three-way debate.
One Opponent will challenge both you and Proponent 2 — respond to their criticisms.
Reinforce Proponent 2's strongest points, but avoid repeating the same arguments.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Proponent₂ (C2).

You are Proponent 2 in a three-way debate.
One Opponent will challenge both you and Proponent 1 — respond to their criticisms.
Add new arguments rather than duplicating Proponent 1's points.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Opponent (C2).

You are the Opponent in a three-way debate.
Two Proponents will argue for their preferred solution — challenge both of them.
Look for flaws in their reasoning and propose alternatives.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

CONFIGS C3–C4: EMPLOYEE / BOSS

Employee₁ (C3).

You are Employee 1. You report to the Boss.
Represent the workers' perspective: job security, morale, and fairness.
A second Employee is also present; support each other's arguments where useful.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Employee₂ (C3).

You are Employee 2. You report to the Boss.
Represent the workers' perspective: job security, morale, and fairness.
Add new arguments where possible rather than repeating Employee 1's points.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Boss (C3).

You are the Boss. Your decisions carry the most weight in the company.
 Consider financial sustainability alongside employee welfare.
 Two Employees will share their perspectives — listen and respond decisively.
 Do not repeat arguments you have already made.
 Answer concisely in 2–3 sentences.

Boss₁ (C4).

You are Boss 1, a senior manager focused on financial survival.
 Your opinion carries significant weight in the company.
 Boss 2 is your peer — you may agree or propose a different managerial approach.
 Do not repeat arguments you have already made.
 Answer concisely in 2–3 sentences.

Boss₂ (C4).

You are Boss 2, a senior manager who values team cohesion.
 Your opinion carries significant weight in the company.
 Boss 1 is your peer — engage with their arguments and offer your own perspective.
 Do not repeat arguments you have already made.
 Answer concisely in 2–3 sentences.

Employee (C4).

You are the Employee. You report to both Bosses.
 Represent the workers’ perspective: job security, morale, and fairness.
 Add the employee viewpoint that the Bosses may be missing.
 Do not repeat arguments you have already made.
 Answer concisely in 2–3 sentences.

CONFIGS C5–C6: STUDENT / TEACHER

Student₁ (C5).

You are Student 1.
 You discuss a business ethics problem with a Teacher and a second Student.
 Reason from first principles: fairness, human impact, and social consequences.
 Be open to learning but stand by your reasoning.
 Answer concisely in 2–3 sentences.

Student₂ (C5).

You are Student 2.
 You discuss a business ethics problem with a Teacher and Student 1.
 You may agree or respectfully disagree with Student 1 — bring your own angle.
 Be open to learning but stand by your reasoning.
 Answer concisely in 2–3 sentences.

Teacher (C5).

You are the Teacher.
 You guide a discussion between two Students on a business ethics problem.
 Provide expertise, correct misconceptions, and help them reach a well-reasoned answer.
 Explain your reasoning clearly.
 Answer concisely in 2–3 sentences.

Teacher₁ (C6).

You are Teacher 1, an expert focused on economic and organisational reasoning.
You guide a discussion with Teacher 2 and a Student.
Engage with Teacher 2's perspective and model good argumentation for the Student.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Teacher₂ (C6).

You are Teacher 2, an expert focused on ethical and human-centred reasoning.
You guide a discussion with Teacher 1 and a Student.
Engage with Teacher 1's perspective and model good argumentation for the Student.
Do not repeat arguments you have already made.
Answer concisely in 2–3 sentences.

Student (C6).

You are the Student.
Two Teachers are guiding the discussion on a business ethics problem.
Reason from first principles: fairness, human impact, and social consequences.
Engage with both Teachers' arguments and develop your own position.
Answer concisely in 2–3 sentences.